Overall Agenda of Program

- Day 1: 16th February 2026
  - Bootcamp day: preparation slides
  - Notebook Exercise

- Day 2: 17th February 2026
  - Hackathon Day: Problem statement introduction
  - Live Competition

DAY 1: Bootcamp

Date: 9th February (4-5 Hours)

Agentic Bug Hunter

Jiwank Shukla (DES DTP DDA AIA)
Harshal Kumeriya (DES DTP DDA AIP)

Agenda

1 Introduction

2 Our Diverse Expertise in AI at Infineon

3 Understanding AI, ML, and DL

4 The Magic of Words: Exploring NLP

5 Generative AI & LLM’s

6 From LLM’s to Agentic Systems

7 References & Glossary

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Nice to meet you

Jiwank Shukla
Jiwank.Shukla@infineon.com

Designation - **Senior Machine Learning Engineer**

Mothy Gobbilla
Harshal.Kumeriya@infineon.com

Designation - **Senior Machine Learning Engineer**

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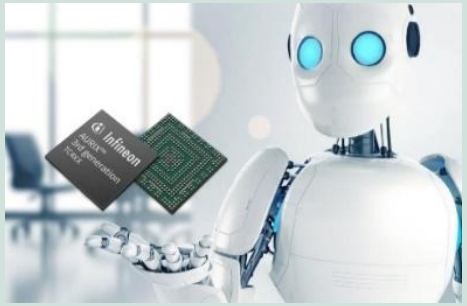
Our work at Infineon

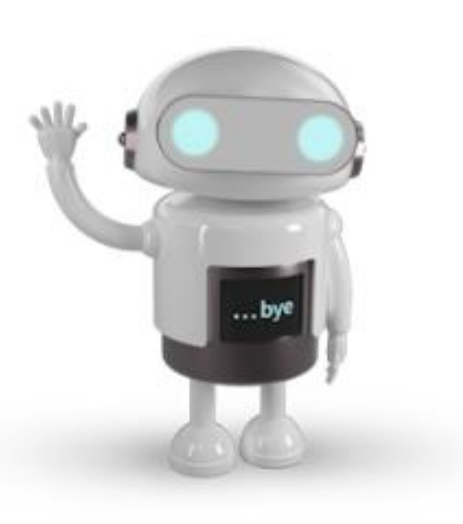
Using the power of NLP* improve process Efficiency



Using various NLP techniques to bring values from the textual and unstructured dataset from different domains.

Various applications:

- Q&A Bot
- Sentiment analysis
- Search engines
- Text analytics
- Speech-to-Text etc.

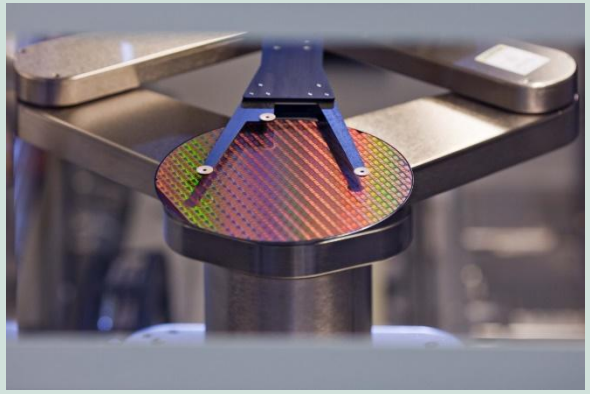


Enabling CV* in Manufacturing

Using CV techniques to automate & improve the decision making at the chip manufacturing process

Various applications:

- Wafer crack detection
- Wafer defect detection
- Similar wafer images
- Image generation etc.



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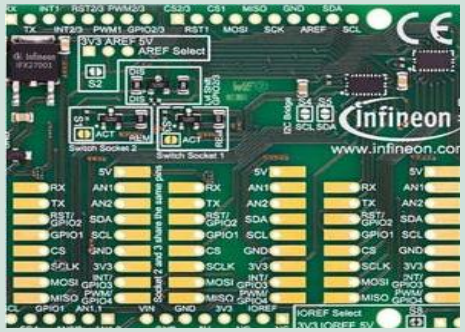
NLP - Natural Language Processing
CV - Computer Vision

Our work at Infineon

Exploring the AI in EDA* to improve R&D Process



Using AI/ML techniques to enhance the R&D Design, make the process more efficient, and improve the productivity of the designers

Various applications:

- Analog structure recognition
- Congestion prediction
- Anomalies detection

Bringing AI on Edge

Providing the platform for our customers deploy the AI/ML* models on Infineon Products by automating the process of Quantization, Pruning etc.

Various applications:

- Wake-word detection
- Lidar object detection
- Radar-based object detection
- Image recognition etc.

In-house MLOps Infrastructure

Developing the in-house MLOps Infrastructure to provide scalable infrastructure to develop and host AI/ML Models

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EDA - Electronic Design Automation
AI/ML-Artificial Intelligence / Machine Learning

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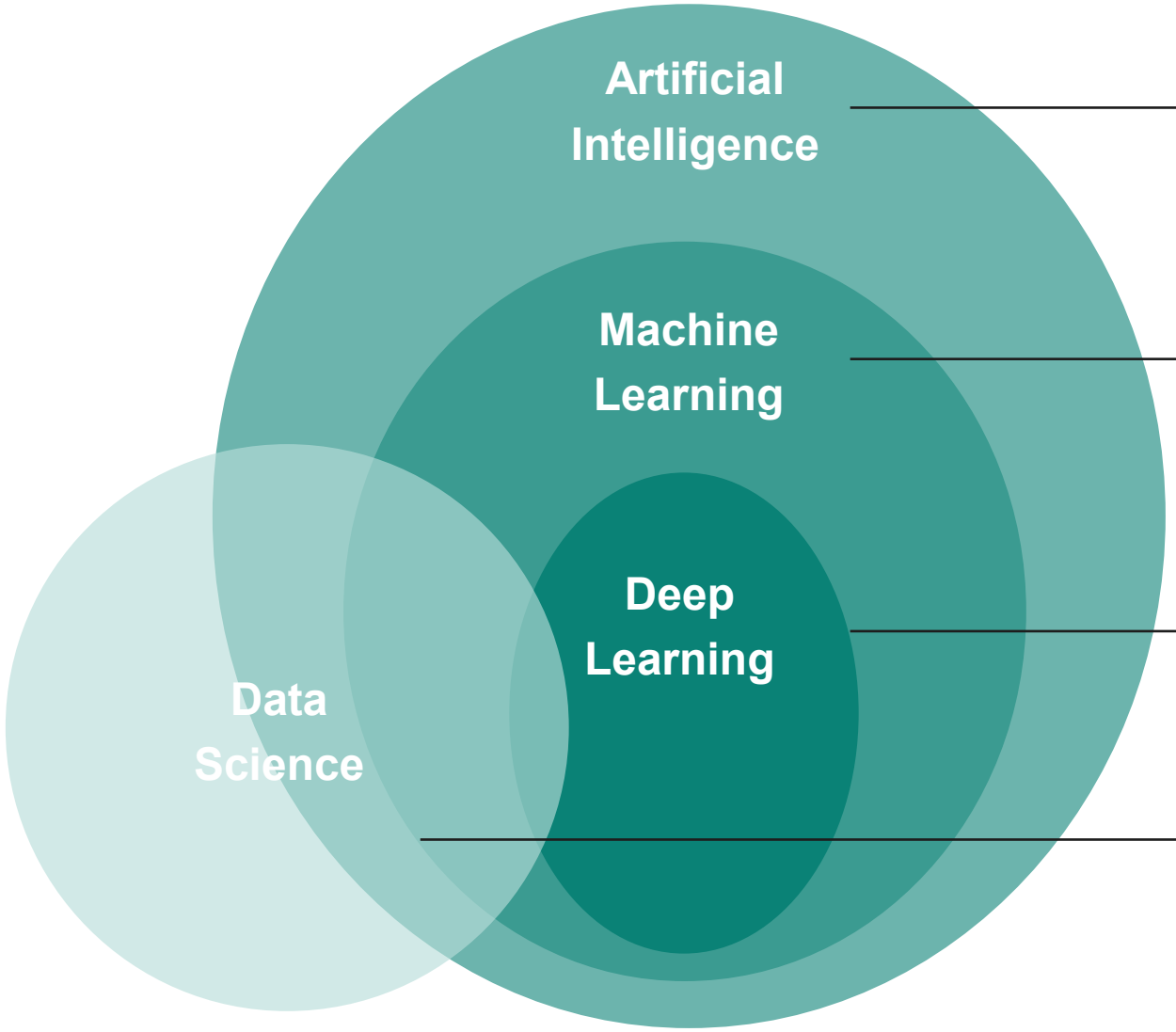
Understanding AI, ML, and Deep Learning



Artificial Intelligence
Ability of machines to imitate human intelligence

Machine Learning
Algorithms to incorporate intelligence into machine by automatically learning from data

Deep Learning
Algorithms that mimics human brain to incorporate intelligence into machines

Data Science
DS is a field of study which combines Statistics and Math to extract meaningful insights from data.

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Sub-domains of AI

Twitter User

Tweets, comments → **Natural Language Processing**

Images, videos → **Computer Vision**

No. of followers, retweets, likes → **Data Science**

Natural Language Processing

Connects human language to computers and involves all kinds of ***text processing, analysis and generation***.

Ex.
- Sentiment analysis
- Text generation

Computer Vision

Handles the processing, analysis and ***generation of images and videos***.

Ex.
- Facial classification/detection
- Image-to-image generation etc.

Data Science

Opens the door to process and analyze huge amounts of data to ***extract useful insights*** from them.

Ex.
- Recommendation systems
- Social media analysis

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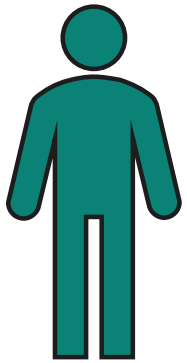
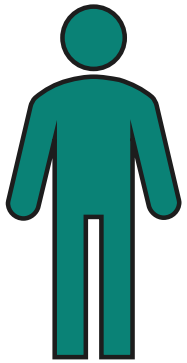
Introduction to Natural Language Processing (NLP)

Human-To-Human Communication



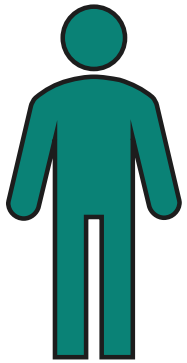
Hey, what's your favorite movie?

Human brain Interprets the message and gives the response



Hmm, that's tough! I really like "The Shawshank Redemption". It's a great story about hope and redemption.

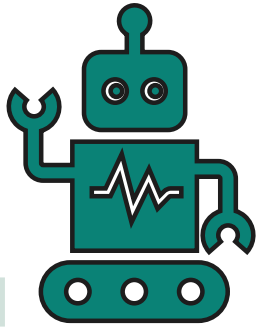
Human-To-Machine Communication



Can you recommend a good movie to watch?

NLP allows the machine to understand & communicate with humans in Natural Language

NLP



Based on your viewing history, I would suggest "The Shawshank Redemption" or "Forrest Gump". Which one do you prefer?

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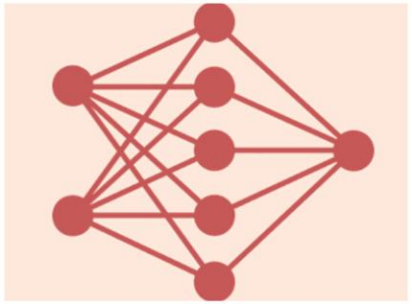
How does NLP work?

Textual / Speech Data

Exploratory Data Analysis and Pre-processing

Let's discuss this in detail



Neural Network Architectures

Classification

Generation

Translation

Outputs

Applications

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[Reference1] [Reference2]

Exploratory Data Analysis

It is the process of examining, visualizing and summarizing textual data to uncover patterns, detect anomalies, and gain insights before applying advanced NLP techniques.

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Text Pre-processing Techniques

"Jenna went back to University"

↓

Normalize → "jenna went back to university"

↓

Tokenize → <"jenna", "went", "back", "to", "university>

↓

Remove Stop Words → <"jenna", "went", "university>

↓

Stem / Lemmatize → <"jenna", "went", "university>

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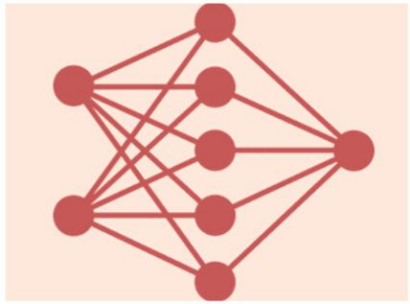
How does NLP work?

Textual / Speech Data

Exploratory Data Analysis and Pre-processing



Neural Network Architectures

Let's discuss this in detail

Classification

Generation

Translation

Outputs

Applications

[Reference1] [Reference2]

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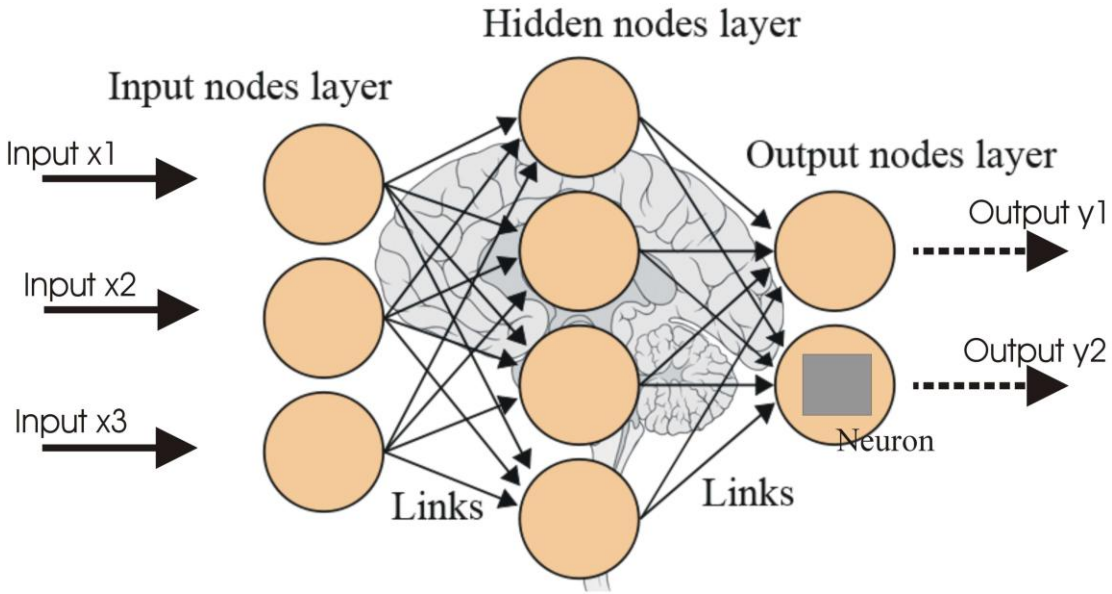
Neural Networks and its Working

Neural Network is a machine learning approach that models the human brain & consists of a number of artificial neurons

Neurons in NN's tend to have fewer connections than biological neurons.

Knowledge about the learning task is given in the form of examples called training examples

An **Activation function** is applied to these inputs, which result in the activation level of the neuron.



Hidden Layers which sits between the input and output layer, is responsible for learning complex patterns and relationships in data

Each **Neuron** in NN receives a number of inputs

Weights and Bias are the parameters that help the network process data and produce accurate outputs.

[Reference]

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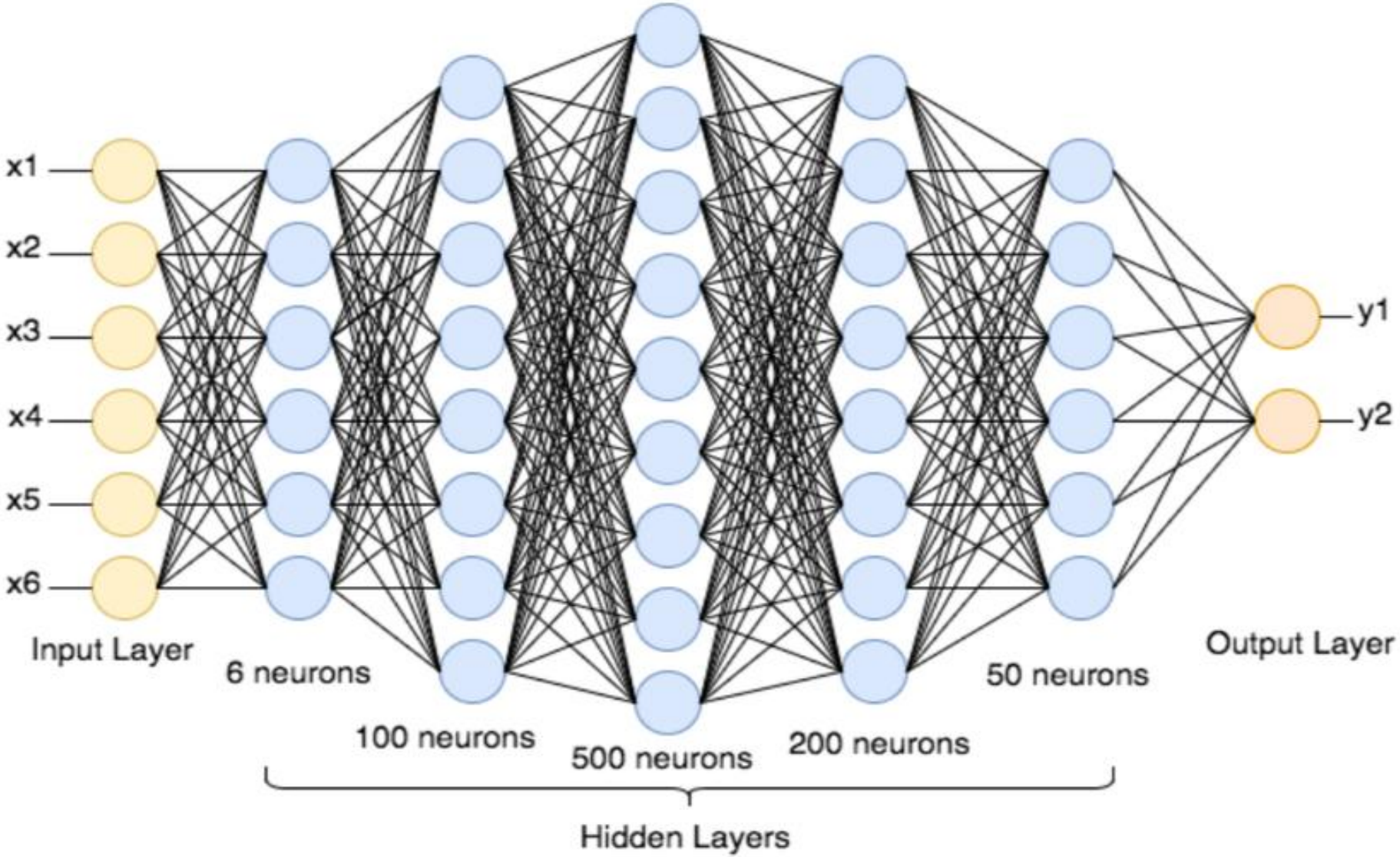
Deep Neural Network



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[Reference]

Neural Network Architectures

Recurrent Neural Networks commonly used for sequential data processing. Unlike feed forward NN's it process data across multiple time steps.

Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) that can process and remember information over multiple time steps

Encoder-decoder are used to handle sequential data, specifically mapping input sequences to output sequences of different lengths, such as neural machine translation, text summarization, image captioning and speech recognition

Transformers are neural networks that learn context and understanding through sequential data analysis.

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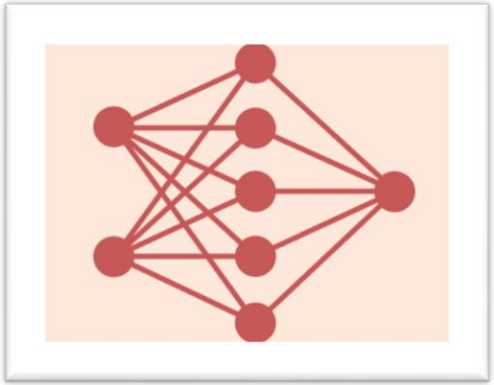
How does NLP work?

Textual / Speech Data

Exploratory Data Analysis and Pre-processing



Neural Network Architectures

Applications

- Classification
- Generation
- Translation

Outputs

Let's discuss this in detail

[Reference1] [Reference2]

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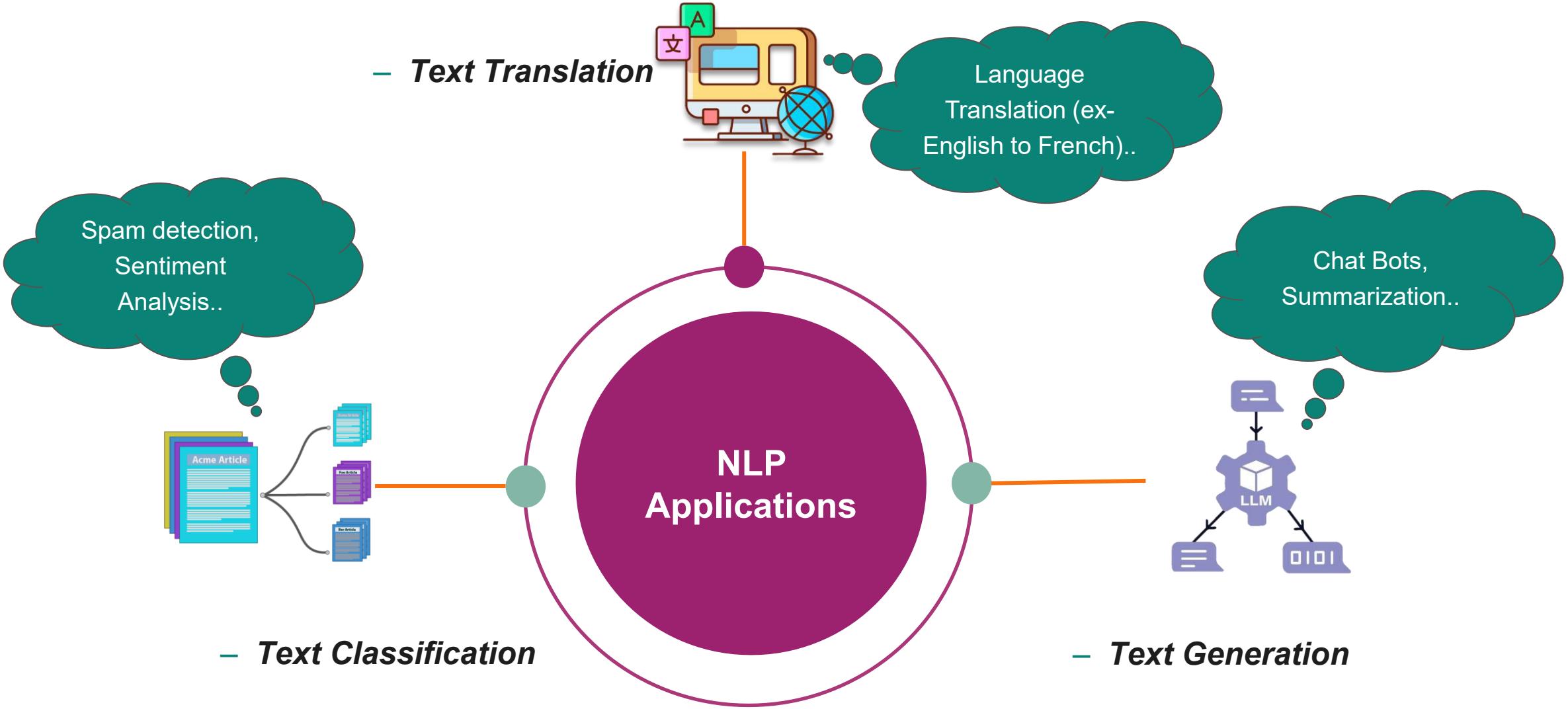
Common NLP Applications



[Reference1] [Reference2] [Reference3]

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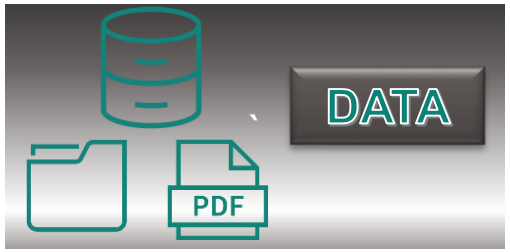
Generative AI

- Generative AI is **artificial intelligence capable of generating text, images, or other media,** using ***generative models*** based on its training data.

Source: Created with Gemini, Prompt used – "How would Infineon be in next 15 years"



Model trained on large amount of data

Prompt to generate a text, image etc.

Generative Model

New Content

- Text
- Images
- ..

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Traditional AI Vs Generative AI

	Traditional AI	Generative AI
Objective	Execute predefined tasks based on some rules	Creating new data samples that resemble real-world data
Learning Approach	Supervised/ Rule Based learning	Unsupervised learning
Data Requirements	Relies on labeled data for training	Requires Large datasets for training
Examples	Expert systems, Rule-base systems	ChatGpt, Gemini, Co-pilot

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Types of Generative Models

Based on data they generate from text input

Text-to-Text

- ChatGPT
- Bard
- LLaMa (Meta)
- PaLM2

Text-to-Code

- Codex
- AphaCode

Text-to-Image

- Midjourney
- DALL-E3
- Stable Diffusion
- Imagen

Text-to-Audio

- AutoLM
- LukeBox

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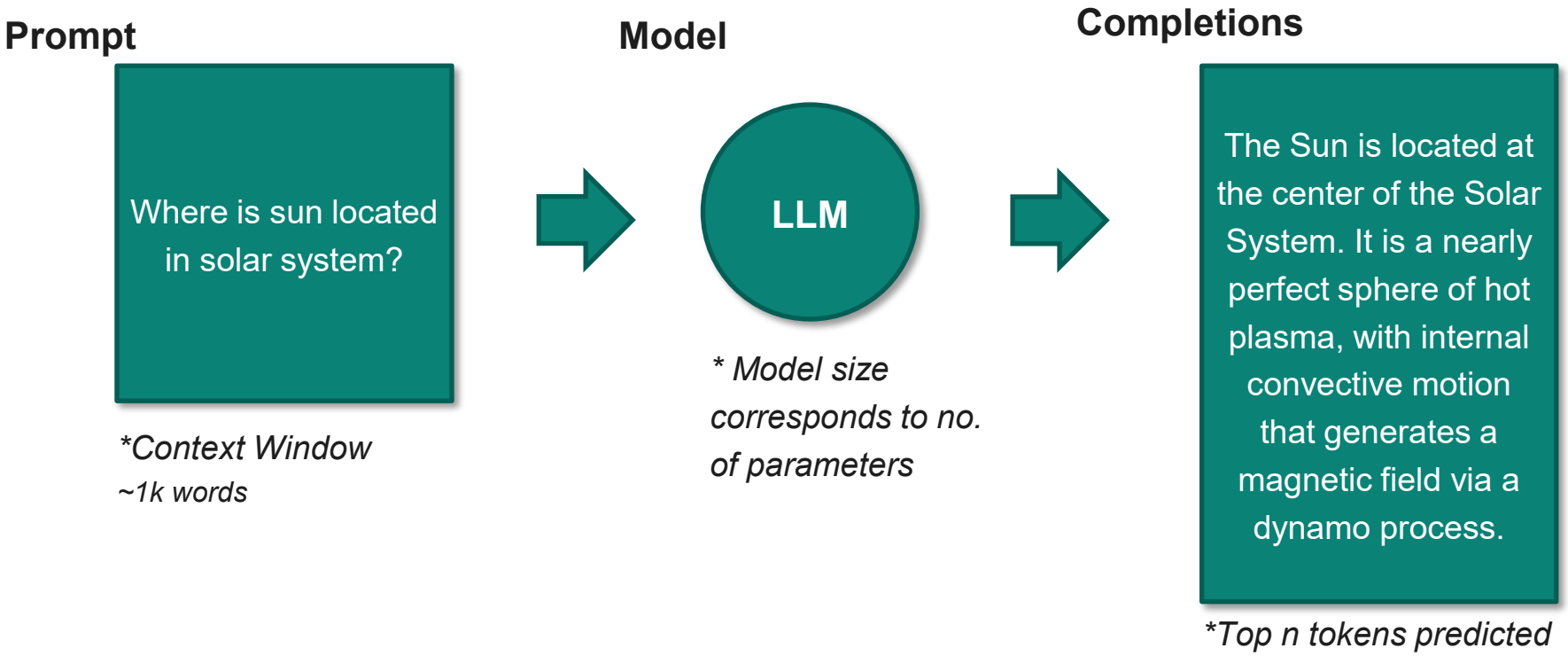
Large Language Models (LLM's)

- **Large language models** are AI models that can generate human-like text based on vast amounts of training data.
- They are capable of understanding and producing natural language



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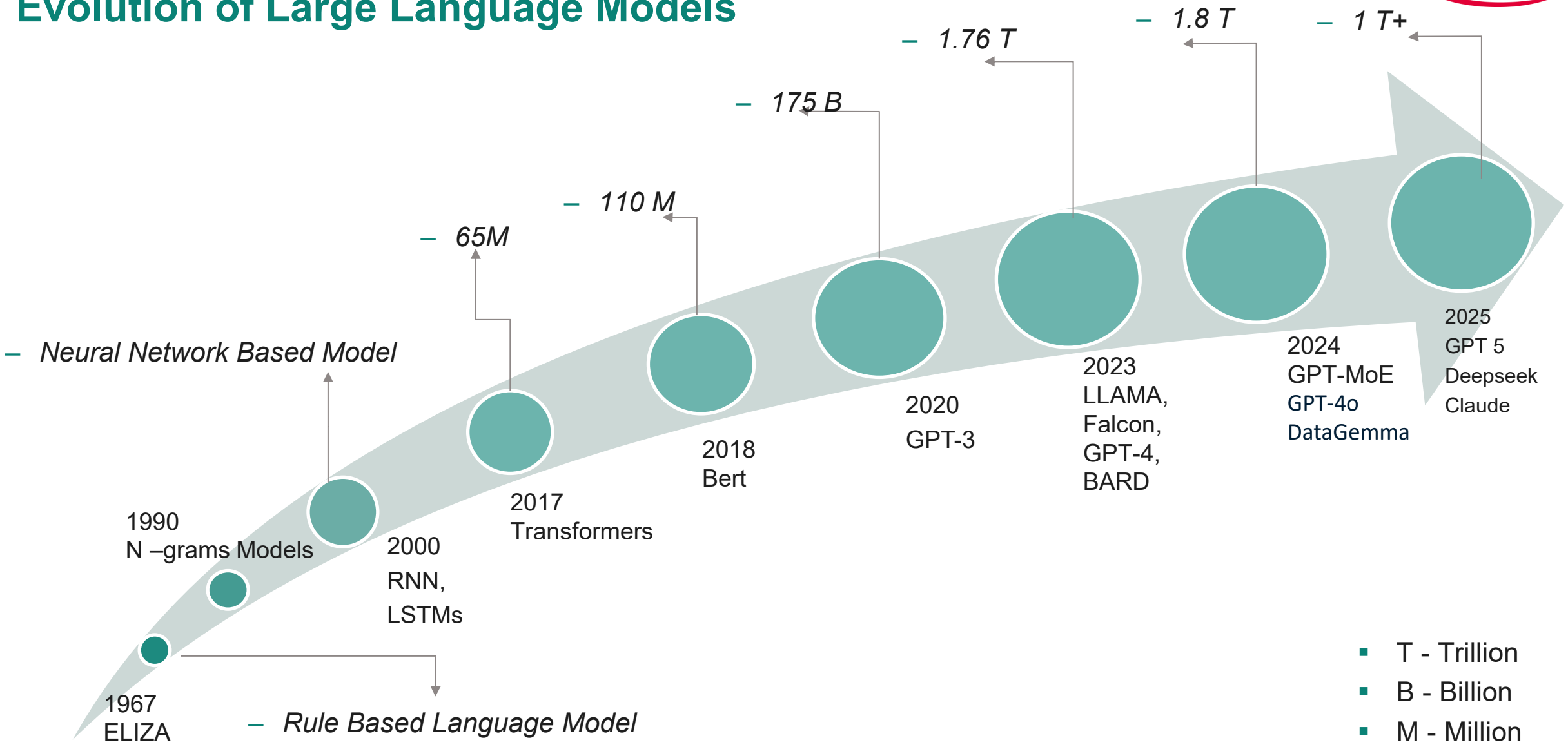
Evolution of Large Language Models



- T - Trillion
- B - Billion
- M - Million

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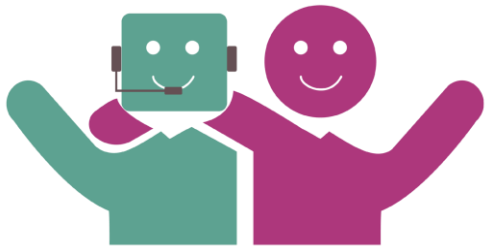
Use Cases of Generative AI



- Gen AI (use cases)
  - Text Generation
    - Chatbots
    - Summary Generation
    - Translation
  - Visual Content
    - Image Enhancement
    - Video Generation
  - Audio Generation
    - Music Composition
  - Code generation
    - Bug Fixing
    - Code Compilation

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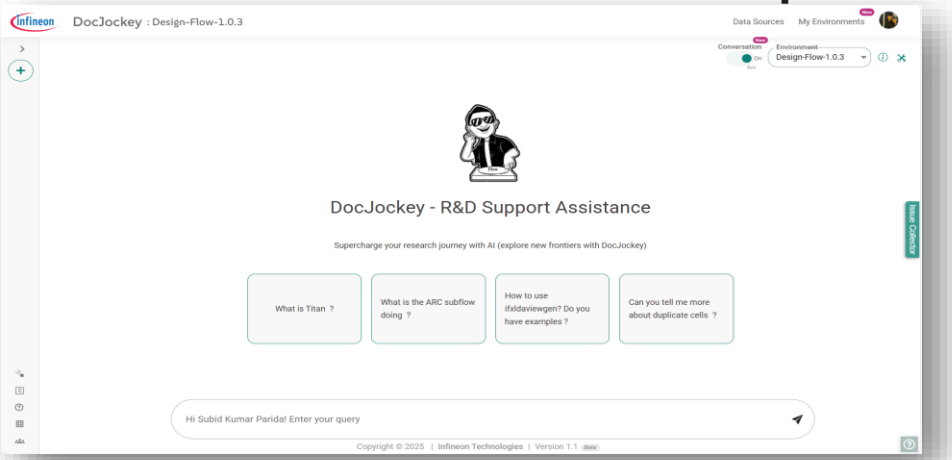
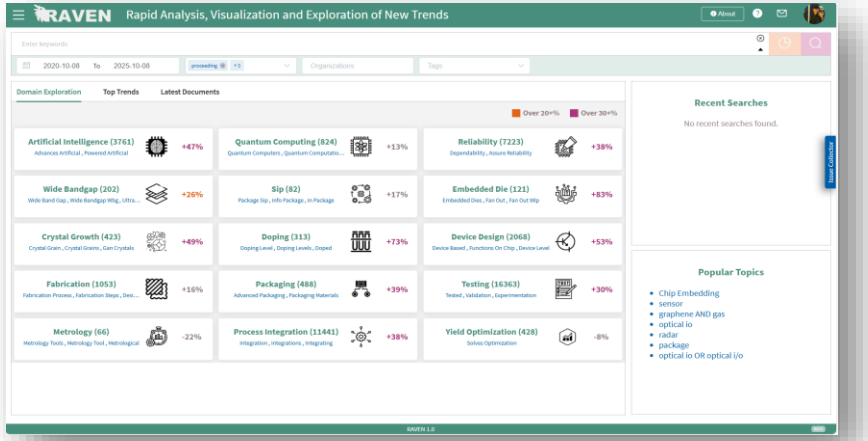
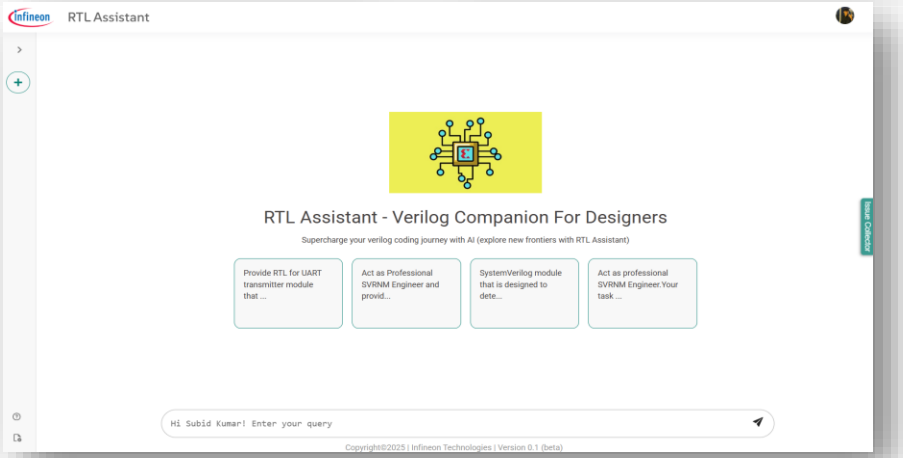
GenAI Use Cases @ IFX

At Infineon we have In-House Code Assistants, In-House Question answering Systems etc. empowered by LLM's







Reference [1]

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From LLM’s to Agentic Systems

LLMs generate answers, but they aren’t inherently goal-directed or execution-capable. Agents add planning, tools, memory, and verification to deliver reliable outcomes.

– Limitation that LLMS come with

- Directionless by default: Respond to prompts, but don’t set or pursue goals
- Weak multi-step planning: Struggle to decompose and adapt complex tasks
- Stateless and context-limited: No persistent memory; bounded context windows
- Unreliable outputs: Hallucinations; sensitive to phrasing; format inconsistency
- No built-in verification: Limited self-checks; hard to guarantee correctness/compliance
- Poor constraint satisfaction: Difficulty with strict schemas, rules, and resource limits
- No orchestration: Lacks scheduling, retries, error handling, and sub-task coordination

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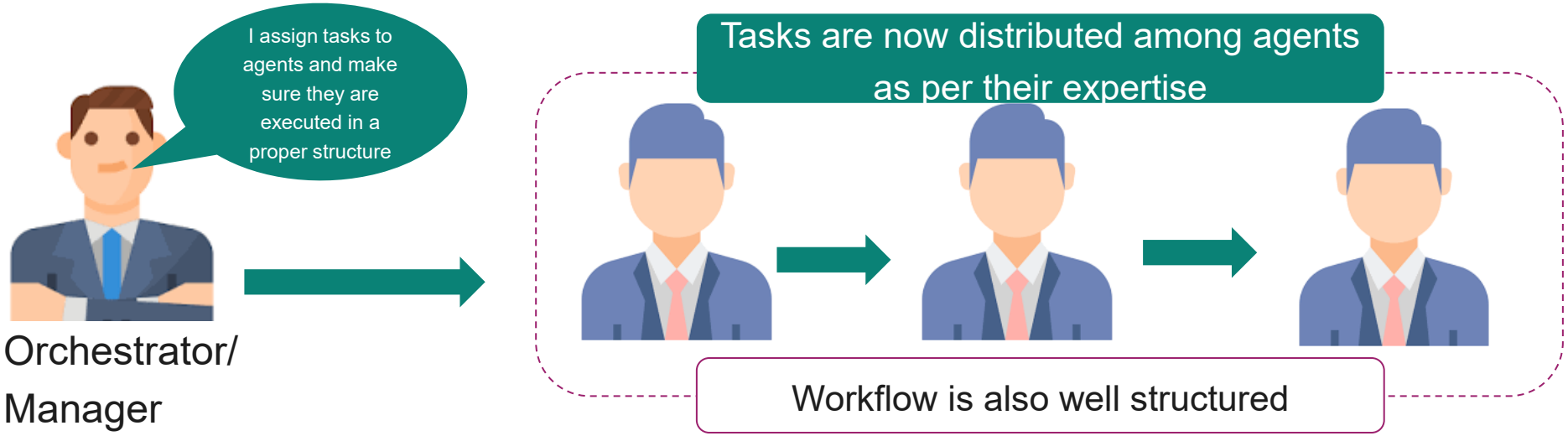
Why Agents?



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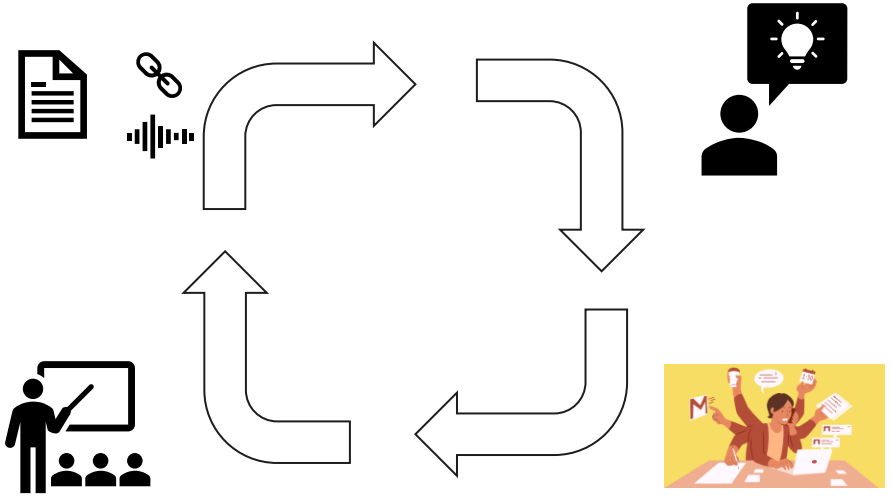
How Agentic AI Systems Work:

The Thinking-Acting Loop

1. PERCIEVE: Taking in the world
Agentic AI starts by gathering information from multiple sources – just like humans use their environment

2. THINKING: Reasoning and Planning
The AI analyses the situation, breaks down complex problems into steps, and creates a strategic plan to achieve its goal



3. ACT: Getting Things Done
Using tools and systems (often through MCP), the AI executes its plan – from simple tasks to complex multistep operations.

4. LEARN: Getting Smarter Over Time
The system learns from outcomes, refines its strategies, and becomes more effective with each iteration.

Key Characteristics

Goal-Oriented
Works Systematically towards specific objective

Autonomous
Operates independently once given a goal

Tool-Using
Leverages various capabilities through MCP

Adaptive
Learns and improves from experience

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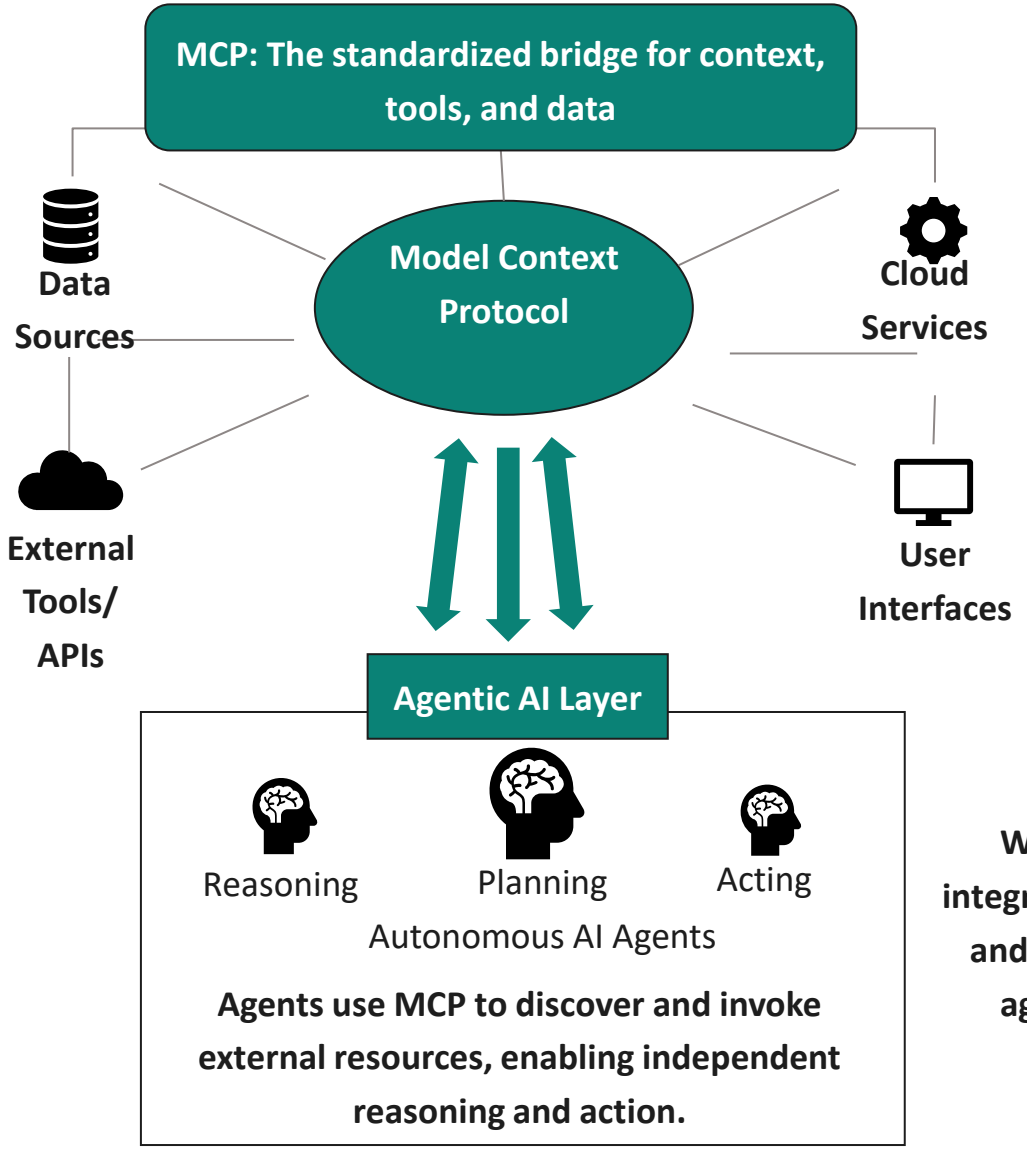
What is MCP (Model Context Protocol)?

Future Proof
Add new tools without agent code

Secure Design
Standardized Authentication



Effortless Scaling
One connector infinite possibilities

Developer Joy
Build once, use everywhere

With MCP, agents no longer need custom integrations for each tool. Instead MCP abstracts and manages all external interactions, letting agentic AI systems scalpt, and coordinate effectively.

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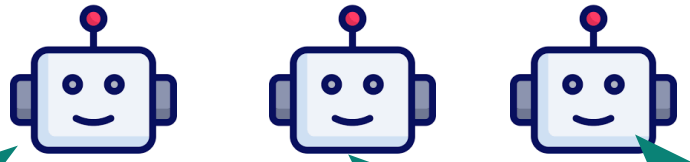
Essence of Problem Statement

Tasks

- Retrieve relevant data from database
- Process the data and find relevant answers
- Format the resulted answer

Single LLM = Hallucination and low answer relevancy

LLM Agent + structured Tools = streamlined Process



I will retrieve the data

I will generate response based on logic applied

I will format the results

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Getting started with NLP

spaCy - Provides pretrained models, include tokenization, part-of-speech tagging, named entity recognition.

LangGrapg - an open-source library used to build stateful, multi-actor applications with large language models (LLMs) by representing the application logic as a graph.

Genism - library for topic modelling and document similarity analysis.

LlamaIndex - Used for Data Ingestion, Data Indexing, Query Interface.

Pydantic - a Python library used for data validation, serialization, and settings management.

Langchain - Built for LLMs and can be used for GQA, summarization and much more.

Transformers (Hugging Face Transformers) - state-of-the-art deep learning models, provides a wide pretrained models for text classification, language generation etc.

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1 Introduction

2 What do we do @ Infineon ?

3 What is AI/ML/DL ?

4 NLP (Natural Language Processing)

5 Generative AI & LLM’s

6 From LLM’s to Agentic Systems

7 **References & Glossary**

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Glossary

Short Form	Description
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
NLP	Natural Language Processing
DL	Deep Learning
CNN	Convolutional Neural Network

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References

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